Fiscal Year:	FY 2006	**Task Last Updated:**	FY 02/02/2006
PI Name:	Flemings, Merton C. Sc.D.		
Project Title:	The Role of Convection and Growth Competition on Phase Selection in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139-4307	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-05
Start Date:	05/01/2000	**End Date:**	11/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	2
No. of Master's Candidates:	5	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Rathz, Tom	**Contact Phone:**	256 544-1409
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Matson, Douglas (Tufts University) Hyers, Robert (University of Massachusetts, Amherst)		
Grant/Contract No.:	NAG8-1685		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Undercooling in containerless processing has shown promise in producing metastable phases and materials that exhibit improved chemical homogeneity and ultra-fine grain sizes. Stainless steel alloys require special handling in order to achieve a significant undercooling range. In particular, chromium is particularly susceptible to contamination through reaction with crucible material. When attempting to perform rapid solidification tests, the formation of active heterogeneous nucleation sites limits the undercooling that may be achieved. By using a containerless technique, this contamination is minimized. The focus of this program is to study of the effect of microgravity on metastable solidification. By observing changes in the mechanism for nucleation of the stable phase following primary recalescence from undercooled melts, the role of convection in phase selection may be evaluated. The value of this research is that the results have application to the design of industrial welding, spray forming and strip casting operations for a commercially important class of structural materials. In addition, this research addresses fundamental issues relating to		

	rapid solidification behavior, metastable phase selection and analysis of the processes governing microstructural evolution. Experiments focus on the use of convection as a processing control parameter.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Control of microstructure during casting, continuous casting, and welding
Task Progress:	Completed modelling of dendrite collision under convective flow with the publication of a new rapid solidification model. Completed ground-base NASA/ESL testing. Continuing with development of RDR presentation for international collaborators and facility design.
Bibliography Type:	Description: (Last Updated: 03/16/2017)
Articles in Peer-reviewed Journals	Fair DJ, Venkatesh R, Boghosian B, Matson DM. "Role of sample size in the nucleation kinetics of phase transformations in steel alloys." Microgravity - Science and Technology. 2005 Mar;16(1):55-8. http://dx.doi.org/10.1007/BF02945946 , Mar-2005
Articles in Peer-reviewed Journals	Matson DM, Hyers RW. "Adiabatic remelting of the mushy-zone during rapid solidification." Philosphical Magazine. 2006;86(24):3795-807. http://dx.doi.org/10.1080/14786430500228150 , Dec-2006
Papers from Meeting Proceedings	Hanlon AB, Hyers RW, Matson DM. "Modeling the effects of internal convection on dendritic evolution in stainless steel alloys." EPD Congress 2005, San Francisco, CA, February 13-17, 2005. In: EPD Congress 2005 : proceedings of sessions and symposia sponsored by the Extraction and Processing Division (EPD) of TMS (the Minerals, Metals & Materials Society) held during the 2005 TMS Annual Meeting in San Francisco, California, USA, February 13-17, 2005. M.E. Schlesinger, Ed. Warrendale, Pa. : TMS, 2005. p. 501-507. , Feb-2005
Papers from Meeting Proceedings	Burke JM, Mueller AJ, Fair DJ, Matson DM. "Application of EM Levitation Processing: Fundamental Studies of Steel Rapid Solidification." Materials Science & Technology Conference, Pittsburgh, Pennsylvania, September 25-28, 2005. In: Proceedings from the Materials Science & Technology Conference (September 25-28, 2005, Pittsburgh, Pennsylvania). Warrendale PA : TMS press, 2005. [CDROM] p. 71-78. ISBN: 978-0-87339-645-5. , Aug-2005